

Circular Saw Crosscut Workbench

Original beginner-friendly design, inspired by “*Turn Your Circular Saw Into a Miter Saw*” (The Honest Carpenter) — built for a 6-1/2" circular saw

Finished size	72"L x 36"W x 35"H	Skill level	Beginner
Saw	6-1/2" circular saw	Joinery	Butt joints, screws + glue — no table saw/miter saw needed

1 SHOPPING LIST

Lumber

Item	Qty	Notes
2x4x8 pine/spruce stud	9	Buy straight boards — sight down each one, reject bowed/twisted stock
3/4" plywood, 4x8 sheet	2	Sheet 1 = main top. Sheet 2 = replaceable cutting panel + 1 spare
1/2" OSB or plywood, 4x8 sheet	1	Lower shelf, cut to 66" x 30"
1x3x8 pine	1	T-square crosscut guide arm
1x4x8 pine	1	Rear workbench fence
1x3x8 pine (or straight MDF strip)	1	Fence for the 8-ft zero-clearance guide
1/4" hardboard (Masonite), 4x8 sheet	1	Only need a 12"x96" strip — rip it down, save the rest
3/4" plywood scrap	—	Free from top/panel offcuts — guide head + triangle brace

Hardware & consumables

Item	Qty	Notes
3" construction/deck screws	1 box (~5 lb)	Frame assembly
1-1/4" wood screws	1 box (~1 lb)	Top, panel, jig parts
Wood glue	1 bottle	T-square guide only — top and cutting panel are screwed, NOT glued
Countersink/pilot bit	1	Prevents split wood near edges

Tools you still need (beyond your circular saw and drill)

Item	Qty	Why
Bar or pipe clamps, 24"–36"	4	Clamp the guide AND the workpiece — required, every cut
Quick-grip style clamps	2	Fast second hand for small pieces and the stop block
Speed square or framing square	1	Squaring the frame and the T-square guide

Tape measure	1	—
Pencil / combo square	1	Marking cut lines
Safety glasses	1 pair	Required, every cut
Hearing protection	1 pair	Required, every cut

Estimated cost

Item	Qty	Est. unit	Est. total
2x4x8 studs	9	~\$4.50	~\$40
3/4" plywood sheets	2	~\$65	~\$130
1/2" OSB (shelf)	1	~\$28	~\$28
1x3x8 pine	2	~\$8	~\$16
1x4x8 pine	1	~\$10	~\$10
1/4" hardboard sheet	1	~\$22	~\$22
Screws, glue, countersink bit	—	—	~\$35
Clamps (4 bar + 2 quick-grip)	6	avg ~\$14	~\$85
Square, safety glasses, ear protection	—	—	~\$35
TOTAL ESTIMATED COST			~\$400

Prices are U.S. home-center ballparks — check actual pricing at your Farr West/Ogden-area store before buying. Total drops to roughly **\$260** if you already own clamps and a square.

2 CUT LIST

2x4 frame — buy 9 boards to allow for kerf loss, defects, and picking straight stock

Piece	Qty	Length	Use
Upper front/back rails	2	72"	Top perimeter, long sides
Upper end rails	2	33"	Top perimeter, short sides
Upper cross supports	3	33"	Under the top, evenly spaced
Legs	4	34-1/4"	Table height 35" minus 3/4" top thickness
Lower rails	2	~66"	Shelf frame, long sides
Lower cross supports	3	~30"	Shelf frame, spaced supports

Lower rail/cross-support lengths are “approximately” because they depend on your actual leg spacing — confirm exact inside measurements once the legs are up (Step 4 of assembly).

Plywood

Piece	Qty	Size	From
Main top	1	72" x 36"	Sheet 1
Replaceable cutting panel	1	24" x 36"	Sheet 2
Spare cutting panel	1	24" x 36"	Sheet 2 (cut both now, store the spare)
Lower shelf	1	66" x 30"	1/2" OSB sheet
T-square guide head	1	4" x 12"	Offcut from Sheet 1 or 2
T-square triangle brace	1	~8" x 8" right triangle	Offcut

1x stock

Piece	Qty	Length	Use
T-square guide arm	1	24"	Rides the saw shoe
Rear workbench fence	1	72"	Back edge of the main top
Stop block	1	~6"	Cut from fence offcut
Zero-clearance guide fence	1	96" (8 ft)	Straight, no crown — pick this board carefully

Hardboard

Piece	Qty	Size	Use
Zero-clearance guide base	1	12" x 96"	Rip from the 4x8 hardboard sheet

3 HOW THE TABLE GOES TOGETHER — ASSEMBLY STEPS

Read Section 9 (Safety Rules) before you cut anything.

Step 1 — Cut all 2x4 frame pieces

Cut the 4 legs, 2 upper rails (72"), 2 upper end rails (33"), and 3 upper cross supports (33") to length. Label each piece in pencil as you cut it ("leg 1," "front rail") so nothing gets mixed up.

Step 2 — Build the upper frame (72" x 36" rectangle)

Lay the 2 long rails (72") and 2 short rails (33") on a flat floor in a rectangle, long rails on the outside at the corners (outside frame dimension becomes 72" x 36-1/2" — the extra 1/2" is rail thickness, expected). Square the rectangle using the 3-4-5 method (Section 6) before driving any screws. At each corner, drive two 3" construction screws through the long rail into the end of the short rail — pre-drill/countersink to avoid splitting. Space the 3 upper cross supports evenly inside the rectangle (roughly every 18"), parallel to the short rails, and screw through the long rails into each end — 2 screws per end.

Step 3 — Attach the legs

Stand the frame up (or flip it and work upside down on sawhorses). Position each 34-1/4" leg inside a corner, flush with the top of the frame. Clamp it, check it's plumb with a square, then drive 3 construction screws through the frame rails into the top of each leg — 2 through one rail face, 1 through the adjacent face.

Step 4 — Build and attach the lower shelf frame

With the table standing, measure the actual inside distance between the legs (front-to-back and side-to-side) about 8" up from the floor — this confirms your ~66" and ~30" lower rail lengths. Cut the 2 lower rails and 3 lower cross supports to those confirmed lengths. Assemble the same way as the upper frame (Step 2), then screw it to the inside faces of the legs, 8" above the floor, using 2 screws per connection.

Step 5 — Check the whole frame for square

Before skinning it with plywood, re-check the top of the frame is square (Section 6) and all 4 legs sit flat without rocking. Fix now — it only gets harder after the top goes on.

Step 6 — Install the main plywood top

Center the 72" x 36" plywood top on the frame. Screw it down with 1-1/4" wood screws every 8"–10" around the perimeter and into each cross support — **no glue**, so it can be replaced later.

Step 7 — Install the replaceable cutting panel

On the right side of the table, mark a 24" x 36" area where the cutting panel will sit. **Beginner recommendation:** simply screw the 24" x 36" panel down directly on top of the main top in that zone (adds 3/4" of height in that one spot, but is far easier for a first build and still fully replaceable). Screw around the perimeter with 1-1/4" screws — no glue. (Advanced option: recess it flush by cutting that section out of the main top first.)

Step 8 — Attach the rear fence

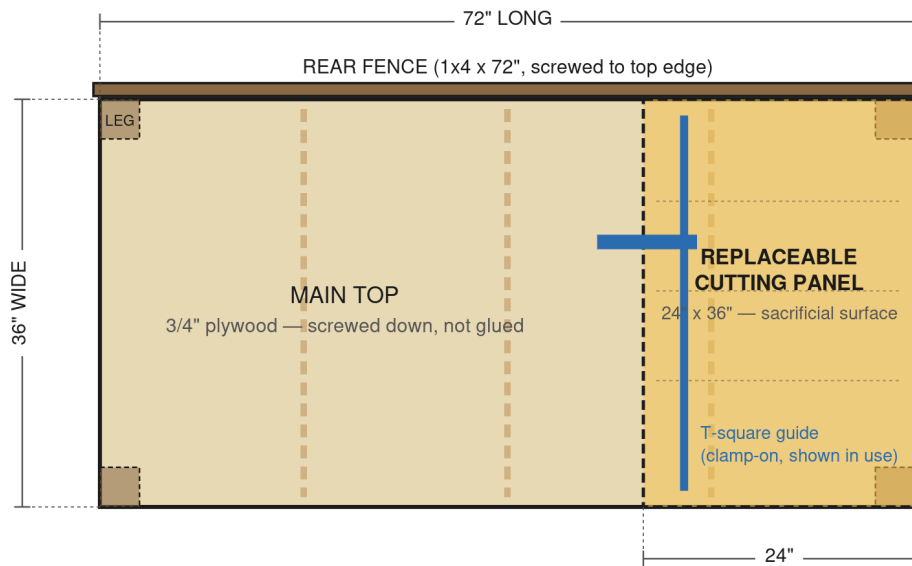
Screw the 1x4x72" fence flat along the back edge of the main top, on top of the plywood, square to the side edges. This is what you'll butt boards against for reference cuts and to register the zero-clearance guide.

Step 9 — Sand and ease edges

Break every sharp plywood and 2x4 edge with 100–120 grit sandpaper. No finish is required for a shop table, but a coat of polyurethane or wipe-on oil on the top makes it easier to wipe clean and slide material across.

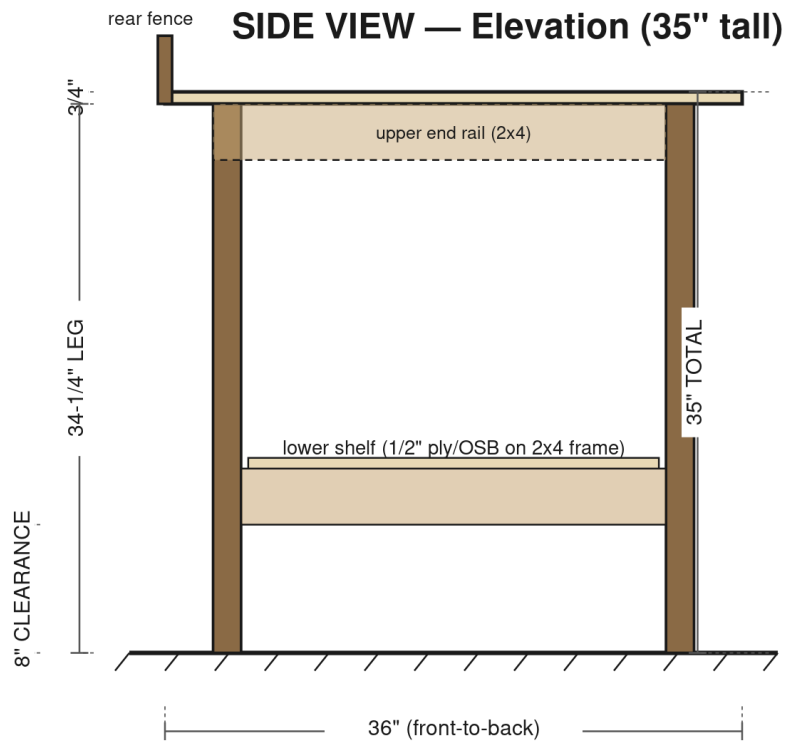
Top View — Main Worktable

TOP VIEW — Main Worktable (72" x 36")



Dashed tan squares = legs below (hidden). Dashed vertical bands = internal cross supports (hidden). Looking straight down at the finished top.

Side View — Elevation



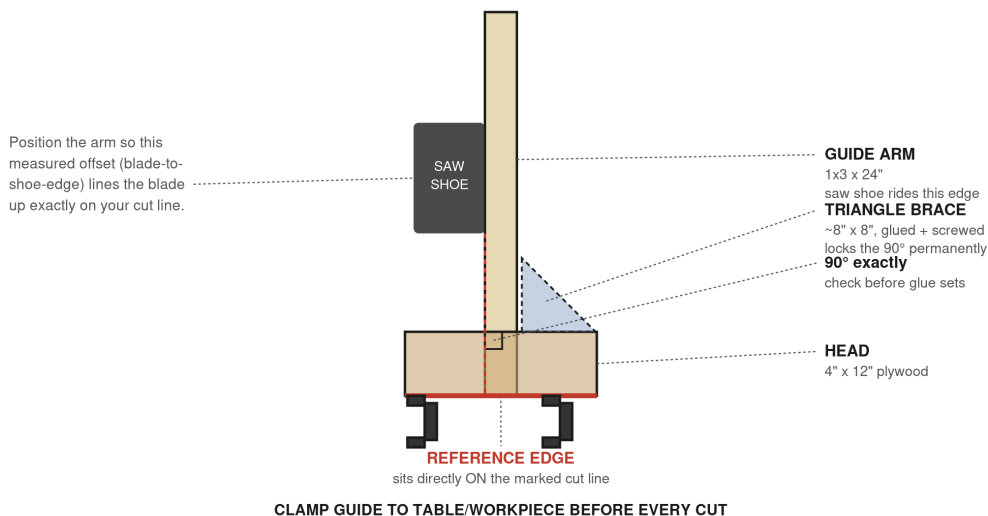
Legs and rails shown at reduced width for clarity. Cross supports and far-side leg are hidden behind the near leg in this view.

4 T-SQUARE CROSSCUT GUIDE — BUILD STEPS

This is the tool that turns your circular saw into a reliable crosscut/miter-style cutter. It's bottomless (no base under the saw shoe), which preserves full cutting depth for your 6-1/2" saw.

- 1 Cut the 1x3 guide arm to 24".
- 2 Cut the plywood head to 4" x 12".
- 3 Glue and screw the guide arm to the head so the arm crosses the head at **exactly 90°**, centered on the head's 12" length, using the head's straight factory edge as your reference — this edge is what registers against your workpiece.
- 4 Check square with a speed square in at least 2 spots before the glue sets; adjust now, not after.
- 5 Cut the triangular plywood brace (~8" x 8" right triangle) and glue/screw it into the inside corner where the arm meets the head, on the back face, to lock the 90° angle permanently.
- 6 Let the glue cure fully (follow the bottle's time) before using the guide for real cuts.

T-SQUARE CROSSCUT GUIDE (top view)



Bottomless design — no base under the saw shoe, so full blade depth is preserved for a 5-1/2", 6-1/2", or 7-1/4" circular saw.

Calibration to your specific saw — do this before your first real cut:

- 1 Measure the distance from the edge of your saw's blade to the edge of the saw's shoe (baseplate) — this is your **offset**.
- 2 Clamp a scrap board to the table with the guide head's reference edge on your marked cut line.

- 3 Position the guide arm so the saw shoe rides against it with the blade lined up exactly on your cut line, using the offset you measured — the guide arm's edge should sit that offset distance away from your intended cut line.
- 4 Make a test cut on scrap. Check the result against your line.
- 5 Once it's dead-on, mark the guide arm itself (paint pen or permanent marker) so you can re-set this same offset instantly next time.

Your saw's max cut depth at 90° with a 6-1/2" blade is typically around 2-1/8"–2-3/8" depending on the exact model — check your saw's manual. That's plenty for the 2x4 frame and plywood in this plan, but it means you can't cut through doubled-up or very thick stock in one pass.

5 8-FOOT ZERO-CLEARANCE GUIDE — BUILD STEPS

Use this one for long rips and long crosscuts on full sheets of plywood, where the T-square guide is too short.

- 1 Rip the hardboard sheet down to a 12" x 96" strip.
- 2 Attach the straight 8-ft 1x3 (or MDF) fence along one long edge of the hardboard strip using glue and 1-1/4" screws every 8"–10", fence flush with or slightly proud of the hardboard edge.
- 3 With the fence clamped to a piece of scrap, run your circular saw down the hardboard with the saw shoe riding against the fence, trimming off the excess hardboard.
- 4 The freshly trimmed edge of the hardboard is now your exact cutting line — line that edge up directly on your marked cut line for future cuts, no measuring or offset math needed.

6 HOW TO CHECK THE FRAME AND GUIDE ARE SQUARE

3-4-5 method (table frame)

- 1 From one corner, measure 3 feet along one edge and mark it.
- 2 From the same corner, measure 4 feet along the adjacent edge and mark it.
- 3 Measure the diagonal distance between those two marks — if the corner is a true 90°, that distance will be exactly 5 feet.
- 4 If it's off, rack the frame gently until the diagonal reads 5 feet exactly, then clamp and fasten.

Diagonal method (whole frame, fast check)

Measure corner-to-corner both ways across the full rectangle. If both diagonals match, the frame is square. If they don't, the frame is racked toward the longer diagonal — push that side until they equal out.

T-square guide

Use a speed square or combination square against both the arm and the head. Check it in 2–3 places along the joint, not just one, since plywood can have a slight bow.

7 MAKING REPEATABLE CUTS WITH A STOP BLOCK

- 1 Set up your T-square guide (or rear fence) for the cut width/angle you need and clamp it.
- 2 Make one test cut to confirm the setting is correct.
- 3 Clamp the ~6" stop block to the table (or to the rear fence) at the exact point where the next board's end should sit.
- 4 Butt each following board against the stop block, clamp it, and cut — every piece comes out the same length without re-measuring.
- 5 Double-check the stop block is still clamped tight every few cuts — vibration works clamps loose.

8 WHAT'S REMOVABLE / REPLACEABLE — AND WHY

Part	Removable?	Why
Main plywood top	Yes — screwed, not glued	Replace if it gets chewed up over years of use
Cutting panel (24"x36")	Yes — screwed, not glued, spare on hand	Sacrificial surface — swap it as kerfs accumulate
T-square guide	Not fastened to the table	Clamp-on tool, stored separately, usable anywhere in the shop
Zero-clearance guide	Not fastened to the table	Same — portable clamp-on tool
Stop block	Clamp-on	Repositions for different cut lengths
Lower shelf	Screwed to legs	Technically removable but meant to stay — not a wear part
Frame (legs, rails, cross supports)	Permanent	This is the structure — no reason to remove it

9 SAFETY RULES — NON-NEGOTIABLE

- Never mount the circular saw upside down under the table.
- Never disable or tie open the lower blade guard.
- Never lock the trigger in the “on” position.
- Clamp both the material AND the guide before every cut.
- Set blade depth only slightly deeper than the material — material thickness plus about 1/8". For a 2x4 (1.5" actual) that's roughly 1-5/8"; for 3/4" plywood, roughly 7/8".
- Support large panels on both sides of the cut so they don't pinch the blade or fall.
- Do not use this jig to rip narrow 2x4 boards lengthwise — it's built for crosscuts, not rips on narrow stock.
- Keep hands clear of the blade's full path, including behind the cut.
- Let the blade come to a complete stop before lifting the saw off the material.
- Eye and hearing protection, every cut, no exceptions.

Copy-paste image prompt — paste into ChatGPT, Sora, Midjourney, or any image tool to visualize the finished table:

A beginner-friendly wooden workshop workbench, 72 inches long, 36 inches wide, 35 inches tall, built from 2x4 pine frame with a 3/4-inch plywood top, natural unfinished wood tone with visible exposed screw heads at the joints. A slightly raised 24x36 inch plywood panel is visible on the right side of the tabletop as a distinct rectangular section, showing faint saw kerf lines. A straight wooden fence runs along the back edge of the table. A separate small T-shaped wooden crosscut guide tool with a black pistol-grip circular saw resting on top sits on the table surface next to the main structure. An open lower shelf sits about 8 inches off the floor between the legs, holding a couple of tool boxes. Set in a home garage workshop with pegboard tool storage on the wall behind it, natural daylight from a side window, photoreal furniture/product photography style, slight depth of field, clean and uncluttered composition.